

- Chapter 1, the system has no free variables.
- By defn, each of the n columns of A has pivots.
 - By defn, a pivot position of A corresponds to the leading 1 of a row in rref.
 - By Theorem 1 from Chapter 1 and the Row Reduction Algorithm, A is converted into a unique rref.
 - By condition 4 from rref, each leading entry must be 1.
 - Since, each of the n columns have a pivot and each row can have ~~at most~~ exactly 1 pivot, $\therefore$ There are exactly n pivot positions.
 - $\boxed{c \Rightarrow b}$ A has n pivot positions. By defn, a pivot position of a matrix corresponds to the leading 1 of a row in rref. By the Row Reduction Algorithm and Theorem 1 from Chapter 1, A is converted to a unique rref. By condition 4 from rref, the leading entry of a row must always be 1. Since each row must have exactly 1 leading entry at most, $\therefore$ each row of A has a leading entry.
 - By condition 2 from rref, for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it.
 - Since there are exactly n pivot columns, each column of A has exactly 1 pivot and it's easy to see why in rref, the pivots are in the main diagonal, i.e. $a_{ii} = 1, 1 \leq i \leq n$.
 - By condition 3 from rref, for each column having a leading entry, all positions below it must contain 0.
 - By condition 5 from rref, for each column having a leading entry, all positions above it must contain 0.
 - $\therefore$ The rref of A is $I_n \because a_{ij} = 0$, where $1 \leq i, j \leq n$ and $i \neq j$.

- $\boxed{b \Rightarrow a}$ This is True by Theorem 7.
- $\boxed{a \Rightarrow k}$ Given A is an invertible matrix. $\therefore$ There exists a unique matrix A^{-1} of size $n \times n$ where $AA^{-1} = I$. $\boxed{\therefore D = A^{-1}}$
- $\boxed{k \Rightarrow g}$ This statement is True from Prob 24 in Ex-2.1.
- $\boxed{g \Rightarrow a}$ This statement is True by Prob 24 from Ex-2.2
- $\boxed{g \Rightarrow h}$ This statement is True by Theorem 4 in Chapter 1.